

QUALITY_STACK

Quality Stack

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`docker-compose.quality.yml` is a **separate, opt-in** compose file holding SonarQube, Prometheus, Grafana, and the one-shot scanner containers (semgrep/trivy/gitleaks/snyk).

Why separate from the main `docker-compose.yml`?

The platform constitution (`.specify/memory/constitution.md` Principle I — Container-First Architecture) defines the product topology as exactly **two containers**: qbittorrent and download-proxy, plus one optional host process (`webui-bridge.py`). Adding SonarQube, Postgres (for Sonar), Prometheus, Grafana, and four scanner images would silently extend that topology from 2 → 8 runtime services, which would violate the principle.

Keeping the quality services in a **separate compose file** means:

1. The product is still a two-container deploy. `./start.sh` launches only those two.
2. Quality tooling is explicit and scope-limited: `-f docker-compose.yml -f docker-compose.quality.yml` only when the developer wants it.
3. Profiles (`quality`, `run-once`, `observability`, ...) gate each service so nothing starts by accident.
4. The constitution does not need an amendment for quality tooling, since no product service depends on it.

See `docs/SCANNING.md` for scanner usage and `docs/OBSERVABILITY.md` for the Prometheus/Grafana story.

Profiles

Profile	Services	When to use
quality	sonarqube, sonar-db	Continuous quality gate host
sonarqube	sonarqube, sonar-db	Alias for above
run-once	semgrep, trivy, gitleaks, snyk	Scheduled / on-demand scans
semgrep, trivy, gitleaks, snyk	single scanner each	Target one tool
observability	prometheus, grafana	Metric collection + dashboards

Environment variables

```
SONAR_DB_USER=sonar          # default
SONAR_DB_PASSWORD=sonar      # default – override in
production
SONAR_TOKEN=...              # CI upload
SONAR_HOST_URL=http://localhost:9000
SNYK_TOKEN=...               # optional
GRAFANA_USER=admin           # default
GRAFANA_PASSWORD=admin       # default – override in
production
```

Ports

All quality services bind to 127.0.0.1: only so they are not exposed to the LAN:

- 9000 SonarQube UI
- 9090 Prometheus
- 3000 Grafana

Lifecycle

```
# Bring up SonarQube (~2 min cold start)
$COMPOSE_CMD -f docker-compose.yml -f docker-compose.quality.yml \
  --profile sonarqube up -d

# Run observability stack
$COMPOSE_CMD -f docker-compose.yml -f docker-compose.quality.yml \
  --profile observability up -d
```

```
# Stop quality services (product stays running)
```

```
$COMPOSE_CMD -f docker-compose.quality.yml --profile quality down
```

./scripts/scan.sh wraps these lifecycle commands for scanner one-shots.